

Contact

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(LinkedIn)

Top Skills

Linear Algebra

Bioinformatics

Databases

Taha Iqbal

MS, MSE* | University of Pennsylvania | Bioinformatician |
Alzheimer's Disease
Philadelphia, Pennsylvania, United States

Summary

Programmer specializing in Alzheimer's disease research, building scalable bioinformatics pipelines and applying advanced computational methods to analyze complex genomic data.

Experience

University of Pennsylvania School of Medicine

Bioinformatician II

August 2021 - Present (4 years 11 months)

Philadelphia, PA

- Developed and applied reproducible NGS and GWAS analysis workflows for large-scale genomic datasets, enabling variant discovery, cohort-level association analyses, downstream statistical modeling, and robust validation across 50,000+ whole genomes
- Performed large-scale genomic data analyses including variant QC, cohort harmonization, phenotype integration, rare variant aggregation, burden testing support, and longitudinal disease progression analyses using population-scale sequencing datasets
- Built computational workflows for genomic data processing and analysis, including variant extraction, annotation, filtering, genotype harmonization, and preparation of datasets for downstream machine learning and statistical genetics applications
- Optimized high-throughput bioinformatics workflows on AWS (EC2, S3, Glacier) and HPC clusters using parallelized processing strategies, significantly improving scalability and runtime efficiency for terabyte-scale sequencing analyses
- Managed and harmonized large-scale genomic datasets across multiple cohorts, including genome build liftover (hg18/hg19 to GRCh38), metadata

standardization, release validation, and reproducible version-controlled data distribution

- Optimized bioinformatics workflows for execution on AWS and HPC clusters using parallelized and scalable processing strategies for large-scale sequencing analyses
- Collaborated with multidisciplinary teams including computational biologists, statisticians, and investigators to interpret genomic findings, support publication-ready analyses, and develop tailored analytical approaches for ongoing research studies
- Mentored junior researchers and lab members on genomic data analysis, pipeline development, statistical workflows, and reproducible computational research practices

Aperiomics

Clinical Bioinformatician I

November 2020 - June 2021 (8 months)

Sterling, VA, United States

Designed and deployed a NGS pipeline for Human papillomavirus (HPV), enabling accurate identification of Alpha, Beta, and Gamma strains for NIH and X4 Pharmaceuticals

Conducted in-house benchmark by simulating reads and aligning closely related HPV genomes from NCBI, establishing thresholds for strain detection and limit of identification

Aligned sequencing reads from clinical samples (fecal, plasma, urine, CSF, tissue) against a curated database of 40,000 bacterial, viral, and fungal genomes for pathogen identification and generate reports for clinicians

Automated data transfer, processing, and report/variant filter generation, improving turnaround time for results used by doctors and patients

Implement containerized workflows (Docker) and executed pipelines on AWS (S3, EC2, Glacier) and local HPC clusters to ensure scalability, reproducibility, and secure data access

Conducted validation studies for fecal and plasma assays and authored SOP's and QC documentation in compliance with CAP regulations to maintain clinical testing standards

Temple University

Researcher

March 2018 - November 2020 (2 years 9 months)

Greater Philadelphia Area

- Built and maintained a SQL database of protein and DNA sequences (multi-species FASTA) to enable large-scale gene duplication and gene clustering analysis
- Conducted computational analyses of biochemical pathways, examining linear vs. cyclic structures, substrates, and intermediates; identified patterns of decreasing molecular similarity with pathway distance
- Evaluated and applied computational/statistical tools to study the impact of natural selection on metabolic pathway length and interspecies metabolic rate variation
- Developed and deployed custom scripts (Python, R, SQL, Bash) to automate retrieval and analysis of biochemical data from public databases, improving reproducibility and efficiency

Education

University of Pennsylvania

Master of Science in Engineering, Scientific Computing · (August 2024 - July 2026)

Temple University

Master's degree, Bioinformatics

University at Albany, SUNY

Bachelor's degree, Biology

University of Pennsylvania

Certificate, Biomedical informatics